



**Pharm***Achieve*

# HERPES SIMPLEX VIRUS (HSV)

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

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**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

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**BID** Twice Daily

**CNS** Central Nervous System

**HIV** Human Immunodeficiency Virus

**HSV** Herpes Simplex Virus

**IV** Intravenous

**NSAID** Non-Steroidal Anti-Inflammatory Drug

**OTC** Over-The-Counter

**PCR** Polymerase Chain Reaction

**PO** Per Os (Oral)

**QID** Four Times Daily

**SEM** Skin, Eye, Mouth

**SPF** Sun Protection Factor

**STI** Sexually Transmitted Infection

**TID** Three Times Daily

# OBJECTIVES

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- 1 Describe characteristics of herpes simplex virus (HSV)
- 2 Understand the route of transmission
- 3 Identify signs and symptoms of herpes labialis and genital herpes
- 4 Understand non-pharmacologic and pharmacologic treatments for herpes labialis and genital herpes
- 5 Understand the importance of HSV in pregnancy and neonates
- 6 Determine appropriate therapy
- 7 Discuss complications of HSV

# DEFINITION

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- Herpes simplex virus causes **cold sores** and **genital herpes**
- There are two types of HSV:
  - **HSV-1:**
    - Primary cause of **herpes labialis (oral cold sores)**
    - **Transmitted via saliva** through oral-to-oral contact, may also be transmitted through oral-to-genital contact
  - **HSV-2:**
    - Primary cause of **genital herpes**
    - **Transmitted via sexual contact** through contact with genital surfaces, skin, sores, or fluids of an infected individual
- **Asymptomatic viral shedding**
  - Patients can be asymptomatic and shedding the virus, thus spreading the virus unknowingly
- Herpes virus travels via axons to nerve cell bodies in the dorsal root ganglia
- **HSV stays in the ganglia**
  - Lifelong latency and recurrence (cyclic)
  - May be reactivated at any time

# ASSESSMENT

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- **Primary infection:**
  - First episode of HSV infection **with no antibodies** for HSV present
  - May be asymptomatic or present as more severe disease (in orolabial infection: primary HSV gingivostomatitis)
  - **Longer duration (7-18 days) and more severe signs and symptoms**
- **Non-primary infection:**
  - First episode in a patient **who has antibodies** for one type of HSV and acquires the other type of HSV
  - **Shorter duration and less severe signs and symptoms** than primary infections
    - Possibly due to partial protection of antibodies of one HSV against alternate serotype
- **Recurrent Episode:**
  - **Latent HSV reactivated:** Likelihood of recurrence greater with HSV-2
  - Typically, **less severe** than primary or non-primary infections
  - **Prodrome symptoms: tingling, itching, pain prior to symptoms**

# CLINICAL PRESENTATION AND COMPLICATIONS

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## Clinical Presentation

The severity of the symptoms depend on:

- Previous exposure: primary, nonprimary, recurrent
- Type of virus: HSV-1, HSV-2
- Site of infection: oral, genital, ocular, nervous system
- Individuals' immune status: immunocompromised

## Complications

- Herpes Gladiatorum: skin infection
- Herpetic Whitlow: finger infection
- Keratoconjunctivitis: conjunctival infection
- Herpes Encephalitis: can cause neurological effects

# RISK FACTORS

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## Risk factors include:

- Physical contact with infected person through:
  - Kissing
  - Sexual contact
  - Sharing utensils (cold sore)
- Having multiple sexual partners
- Female > Male
- Immunocompromised

## Complications

- Stress/fatigue
- Infection/fever
- Immunosuppression
- Hormonal changes
- Menstruation
- Physical trauma/dental extractions/surgery
- Sun exposure
- Temperature extremes



# HERPES LABIALIS

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- Also known as **cold sores or fever blisters**
- Diagnosis is mainly accomplished by the **clinical presentation of the lesion**
- **No cure** or treatment to eradicate herpes virus
- Lays dormant in ganglia until recurrence
- Begin treatment during prodrome symptoms
- **Takes about 1-2 weeks for cold sores to fully resolve**



# CLINICAL PRESENTATION PRIMARY INFECTION: HERPES LABIALIS

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- First infection referred to as **primary herpes gingivostomatitis**
- Lasts from **1-3 weeks**
- **Commonly asymptomatic**
- Symptomatic patients may present with:
  - Painful blisters around the mouth area
    - blisters may crack, release clear, sticky liquid, then crust over
  - Lymphadenopathy
  - Red, swollen gums
  - Sore throat
  - Systemic: fever, malaise, myalgia

# CLINICAL PRESENTATION RECURRENT INFECTION: HERPES LABIALIS

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- Recurrent infection as latent HSV reactivated
- Lasts about 8-10 days
- Asymptomatic shedding
- Prodrome symptoms: tingling, itching, burning before blisters and sores
- Signs and symptoms:
  - Painful blisters around the mouth area
  - Red and erythematous base

# GOALS OF THERAPY

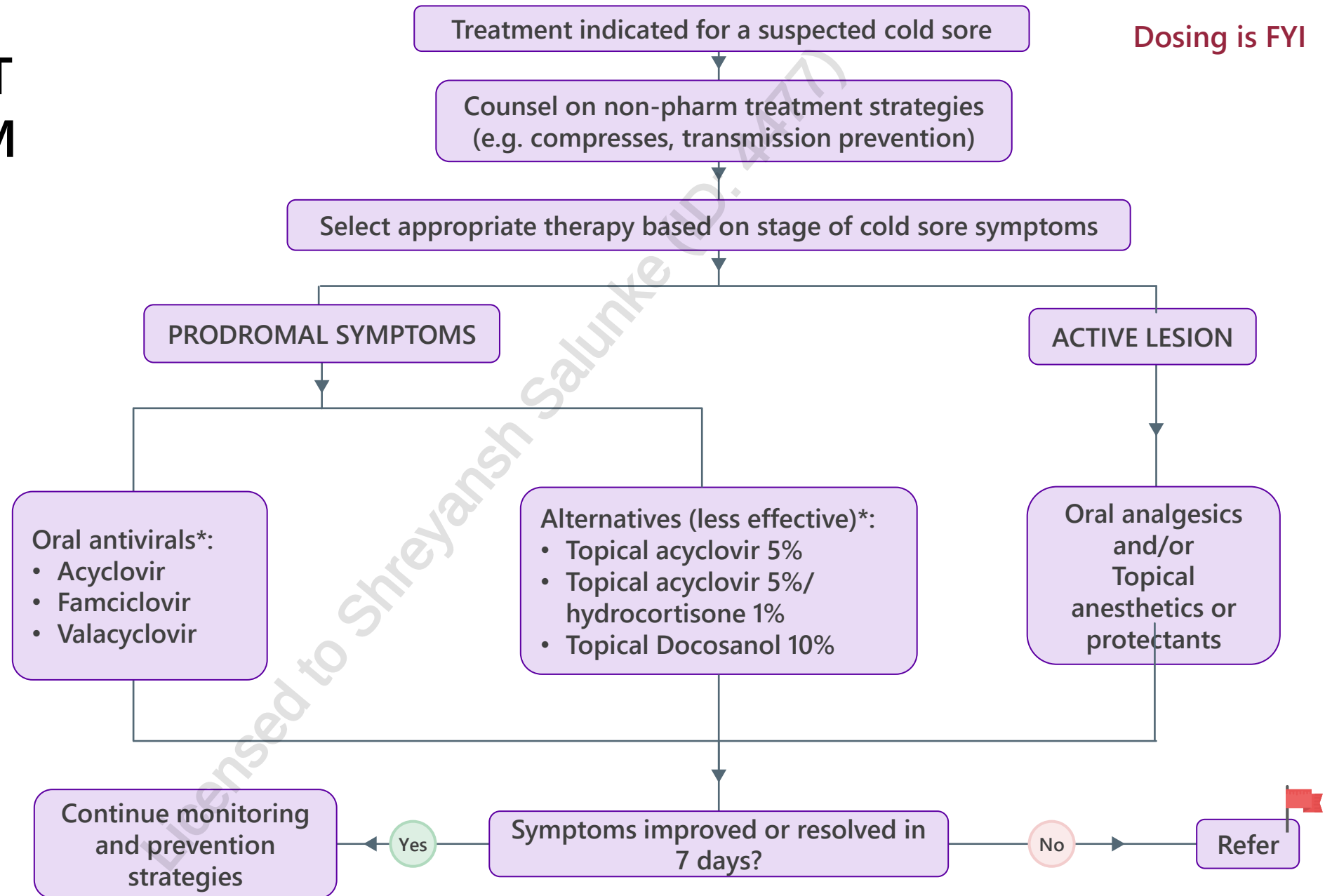
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- ✓ Minimize pain or discomfort
- ✓ Minimize the duration of lesions and viral shedding
- ✓ Prevent autoinoculation
- ✓ Prevent recurrences
- ✓ Prevent transfer of virus

# TREATMENT ALGORITHM

## HERPES LABIALIS (COLD SORES)

Dosing is FYI



\*Treatment options have varying age restrictions; ensure that the treatment prescribed is appropriate for the patient's age

# NON-PHARMACOLOGIC MEASURES: HERPES LABIALIS

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- **Prevention**
  - If stress is a known trigger, reducing stress may reduce occurrences
  - Prevent sun-induced recurrences by using sunscreen with SPF 30+ on affected areas
  - Frequently wash hands to prevent transmission of the virus to other parts of the body or other individuals
  - Avoid skin-to-skin contact until blister has dried up and crusted over
- Gently clean area with mild soap and water
- Cool or warm compresses
- Topical protectants prevent blisters from drying out and breaking
  - Petrolatum
  - Cocoa butter
  - Calamine
- Prevent autoinoculation
  - Avoid oral sex, kissing, and touching lesions until lesions completely healed
  - Frequently wash hands

# PHARMACOLOGICAL MEASURES (OTC)- HERPES LABIALIS

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## Antiviral

- Docosanol cream (Abreva®)
  - Prevents HSV from spreading to healthy cells
  - Apply at the first sign of pain, itching, burning, redness, or tingling
  - May decrease duration of painful symptoms and time to healing by 17 to 79 hours

## Pain relief/Protectants

- Moderate-to-severe pain: Acetaminophen or NSAIDs
- Mild pain: Topical Products
  - Benzocaine (Anbesol®)
  - Lidocaine/Prilocaine (EMLA®)
  - Camphor, menthol (Blistex®, Anbesol Cold Sore®)
  - Lidocaine
  - Pramoxine
  - Zinc sulfate/ Heparin sodium (Lipactin®)

# PHARMACOLOGICAL MEASURES (RX)- HERPES LABIALIS

- Systemic antivirals for treatment of primary infection (Note: uncommon)

Dosing is FYI

| Antiviral               | Adult Dosing           |
|-------------------------|------------------------|
| Acyclovir (Zovirax®)    | 200 mg 5x/day x 7 days |
| Famciclovir (Famvir®)   | 500 mg BID x 7-10 days |
| Valacyclovir (Valtrex®) | 1 g BID x 7-10 days    |

- Systemic antivirals for treatment of recurrent infections

| Antiviral               | Adult Dosing                                        |
|-------------------------|-----------------------------------------------------|
| Acyclovir (Zovirax®)    | 400 mg 5x/day x 5 days                              |
| Famciclovir (Famvir®)   | 750 mg BID x 1 day OR 1500 mg once as a single dose |
| Valacyclovir (Valtrex®) | 2 g BID x 1 day                                     |

- Topical therapy
  - Acyclovir 5% Cream/Ointment 4-6x/day for up to 10 days
  - Acyclovir 5%/Hydrocortisone 1% Cream 5x/day for 5 days
  - Docosanol 10% Cream 5x/day for up to 10 days



# REFERRAL CRITERIA- HERPES LABIALIS

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- Immunocompromised patients
- Patients with recurrent, persistent cold sores ( $\geq 6$ x/year)
- Moderate to severe primary infection (systemic symptoms)
- Lasts longer than 2 weeks
- Signs of secondary bacterial infection

Select patients may benefit from antiviral therapy (most beneficial if started early, within 48 hours of symptom onset)

# GENITAL HERPES

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- Genital herpes is a STI
- Genital herpes increases the risk of contracting HIV
- Recurrence is more likely to occur with HSV-2 infection
- Rate of recurrence decreased with increasing age
- Diagnosis should be confirmed with lab tests to exclude other STIs (e.g. chancroid)
  - PCR
  - Viral culture
  - Serology lab tests

# CLINICAL PRESENTATION PRIMARY INFECTION- GENITAL HERPES

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- May be asymptomatic or unrecognized by the patient
- Viral shedding lasts on average about 12 days
- Lesions heal in 2-4 weeks
- Signs and symptoms of genital herpes include:
  - Painful, pustular lesions
  - Local itching, tingling, pain
  - Dysuria
  - Lymphadenopathy: swollen glands
  - Systemic symptoms: fever, myalgia, malaise

# CLINICAL PRESENTATION- RECURRENT GENITAL HERPES

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- Recurrent infections are generally milder and have a shorter duration than primary infection
- Viral shedding lasts on average about 4 days
- Lesions heal in 7-10 days
- Begin treatment during prodrome symptoms
- Suppressive therapy may be indicated if the patient:
  - Is experiencing frequent recurrences ( $\geq 6$  recurrences per year)
  - Is experiencing complications from infection
  - Is serodifferent or has serodiscordant partners
  - Has multiple sexual partners
- Use of suppressive therapy should be reevaluated on a yearly basis

# THERAPEUTIC ALTERNATIVES- GENITAL HERPES

Dosing is FYI

- All patients with genital herpes should be treated with an antiviral
- Genital herpes in HIV infected/immunosuppressed patients should be treated for a longer course than listed below, consider IV acyclovir

| Drug                      | Primary Episode                                                                                                                | Recurrent Episode                                                                                 | Suppressive Therapy                                                                               |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Acyclovir<br>(Zovirax)    | 200 mg 5x/day for 7 days OR<br>400 mg TID for 7-10 days (US)<br>IV for severe cases<br>Start within 7 days of<br>symptom onset | 200 mg 5x/day for 5 days OR<br>800 mg TID for 2 days<br>Start within 12 hours of<br>symptom onset | 200 mg 3-5x/day OR<br>400 mg BID                                                                  |
| Famciclovir<br>(Famvir)   | 250 mg TID for 5-10 days<br>Start within 5 days of<br>symptom onset                                                            | 125 mg BID for 5 days<br>Start within 6 hours of<br>symptom onset                                 | 250 mg BID                                                                                        |
| Valacyclovir<br>(Valtrex) | 1000 mg BID for 10 days<br>Start within 3 days of<br>symptom onset                                                             | 500 mg BID for 3 days OR<br>1000 mg daily for 3 days<br>Start within 12 hours of<br>symptom onset | 500 mg daily ( $\leq 9$<br>recurrences per year)<br>1000 mg daily ( $>9$<br>recurrences per year) |

# HSV IN PREGNANCY

Dosing is FYI

- **HSV can be transferred to the neonate**
  - **Mainly during labour:** direct contact during virus shedding
  - Rare to transfer transplacentally
- Greatest risk is during a primary infection **near term**
- **Caesarean required** if there's an active infection or prodrome symptoms are present during delivery
- Neonatal HSV may cause skin, eye, mouth (SEM) disease and can advance to CNS or disseminated disease
  - Increased risk of morbidity and mortality
- **Suppressive antiviral management is recommended at 36 weeks**
  - Reduce risk of recurrence
  - Caesarean is not required
  - Limited data on famciclovir

| Drug                      | Pregnancy                   | Neonates     |
|---------------------------|-----------------------------|--------------|
| Acyclovir<br>(Zovirax)    | 200 mg QID OR<br>400 mg TID | IV acyclovir |
| Valacyclovir<br>(Valtrex) | 500 mg BID                  |              |

# CNS HSV

Dosing is FYI

- **HSV encephalitis/meningitis is a medical emergency**
  - Typically in elderly, immunocompromised patients, but can occur in anyone
  - Leads to seizure, personality changes, fever, photophobia
  - HSV-2 meningitis less severe than HSV encephalitis
- Rapid diagnosis and treatment is essential for reducing morbidity and mortality
- Intravenous acyclovir is the drug of choice for majority of patients
- HSV encephalitis should be treated with IV therapy for 14-21 days
- HSV meningitis may be treated with oral therapy for 10-14 days

| Drug                   | Dose               |
|------------------------|--------------------|
| Acyclovir<br>(Zovirax) | 10-15 mg/kg IV Q8H |

# THERAPEUTIC TIPS- GENITAL HERPES

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- Start therapy as soon as possible for maximum benefit
- Acyclovir has a more complicated regimen than famciclovir and valacyclovir but is less expensive
- Lesions may be painful and cause patient to avoid bowel movements
  - Analgesics may be used for the pain
  - Laxatives may help with bowel movements
- Counsel on safe sex and prevention of transmission
  - Use a condom
  - Avoid contact with lesions
  - Avoid sexual activity during prodromal symptoms and healing process of lesions



# CASE 1

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AY is a 29-year-old female who presents with genital itching, pustular lesions, dysuria, and malaise. She has a history significant for genital herpes, which she was diagnosed with about 1 year ago, with HSV-2. She was treated then and has been asymptomatic since that primary infection. AY is a teacher and has been feeling extremely stressed as she has report card deadlines approaching and parent-teacher interviews shortly after. Approximately 8 hours ago, she began experiencing a burning and tingling sensation around the vulvar area and came to your family health team, seeking help. Her past medical history also includes asthma for which she uses fluticasone propionate 250 mcg 1 puff inhaled twice daily, salbutamol 100 mcg 1-2 puffs inhaled every 4 hours as needed, acetaminophen 500 mg PO as needed, and probiotics taken as 1 capsule PO daily. She has no known drug allergies, does not smoke, does not use recreational drugs, but likes to have a few drinks, usually 4-6 drinks over the weekend. All her vitals are stable, and her most recent lab results showed her CrCl to be over 90 mL/min.

1. What factor may increase AY's risk for recurrent genital herpes?
  - a) Asthma
  - b) Alcohol consumption
  - c) Job stress
  - d) Age



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2. What therapeutic alternative is most suitable for AY?

- a) Valacyclovir 500 mg PO BID x 3 days
- b) Famciclovir 250 mg PO TID x 5 days
- c) Valacyclovir 2 g PO BID x 1 day
- d) Acyclovir 400 mg PO BID x 10 days



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3. On what should you counsel AY?
- a) Antiviral therapy may shorten duration of lesions, but does not cure the infection
  - b) Transmission of HSV to a sexual partner is not possible while on antiviral therapy
  - c) The most common side effects of this medication are nausea/vomiting and renal failure
  - d) Antiviral therapy duration is similar for all patients and clinical indications



# CASE 1 - QUESTIONS

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AY is a 29-year-old female who presents with genital itching, pustular lesions, dysuria, and malaise. She has a history significant for genital herpes, which she was diagnosed with about 1 year ago, with HSV-2. She was treated then and has been asymptomatic since that primary infection. AY is a teacher and has been feeling extremely stressed as she has report card deadlines approaching and parent-teacher interviews shortly after. Approximately 8 hours ago, she began experiencing a burning and tingling sensation around the vulvar area and came to your family health team, seeking help. Her past medical history also includes asthma for which she uses fluticasone propionate 250 mcg 1 puff inhaled twice daily, salbutamol 100 mcg 1-2 puffs inhaled every 4 hours as needed, acetaminophen 500 mg PO as needed, and probiotics taken as 1 capsule PO daily. She has no known drug allergies, does not smoke, does not use recreational drugs, but likes to have a few drinks, usually 4-6 drinks over the weekend. All her vitals are stable, and her most recent lab results showed her CrCl to be over 90 mL/min.

4. AY returns 1 year later and informs you that she is 24 weeks pregnant. What would you suggest?
- a) Suppressive therapy with valacyclovir at 36 weeks
  - b) Treatment with acyclovir IV for her at this time
  - c) Consultation for Caesarian section
  - d) Nothing at this point



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# CHANGE LOG

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## March 2025

- Slide 6: Made distinction in second bullet that primary gingivostomatitis is Circle referring to orolabial infection

## July 2025

- Formatting changes throughout; legend and references updated
- Slide 16: Updated doses

## October 2025

- Slide 13: Added the word 'topical' to docosanol

## March 2026

- Slide 2: Updated NAPRA competencies